POST

It ain't just physical" . . .

September 29, 2022

To more effectively protect the environment, we must move past our understanding of the environment as fundamentally and merely physical , , ,

---“Is the world fundamentally physical?”

- Jed Anderson, EnviroAI

We now know that this is what the world more fundamentally “looks” like “below the surface” of things.

Yup. It doesn’t really “look” like anything.

It’s not physical waves.

It’s not physical particles.

IT’S A PROBABILITY WAVE FUNCTION.

What causes the probability wave function to turn into the physical world around us … we don’t know yet.

The way it looks “underneath the surface” doesn’t affect it’s beauty on the surface in my opinion. It’s still beautiful. In fact, I think it’s more beautiful with this deeper understanding.

DISCLAIMER: You cannot blame me if you don't like the way it more fundamentally "looks" "underneath the surface". I didn't make it this way. You can point that finger at God if you find it upsetting. And I didn't discover or prove it was this way. You can point that finger at Einstein, Bohr, Heisnberg, Schroedinger, de Broglie, Dirac, Maxwell, Planck, Bohm, Pauli, Bell, and Clauser. Personally I think it looks cooler. I like it. But it really doesn't matter whether I like it or not. My personal feeling or comfortability with truth doesn't define it. Truth will not adjust its perceptibility to my reality. I must adjust my perception of reality to truth. Truth is truth. It's just deeper truth.

MORE TRUTH IS ALWAYS GOOD

… If you lived in the 1500's, would you tell your kids the world is flat? No.

… If you lived in the 1800's, would you tell your kids the sun travels around the earth? No.

… If you live in the 2020's, would you tell your kids that matter in and around them at the most fundamental level of our understanding is "physical"?

PHYSICS

The continual iconoclastic evolution in our understanding of physics …

- Aristotelian Physics (345 BC)
- Universe is made of 5 elements (https://r20.rs6.net/tn.jsp?f=001uHpNpTz3Ars5Xym_7wmbY9gdRgd5ZFUG92SAQAINR9rQyEN8WJ6RQbW8Uq2cqnS4YQ7r5ap0gnco1ZYdaJYTvnJuXLBjDFPp4Is-NpL4s6JwVUfbYZq-g-D8Y_M0k0E09ao6jv8lR1l5lpFHm-FxSN36wjKwsq

UMQTjKhrdgCWA84R18qy926A==&c=&ch=): earth, air, fire, water, and aether. Objects have natural motions: those of earth and water tend to fall; those of air and fire, to rise.

- Galilean Physics (1592)
- Galileo determines first laws of motion by experimentation
- Develops the concepts of: velocity, force, and inertia
- natural state of motion is uniform motion or rest
- objects fall at same rate regardless of mass
- Keplerian Physics (1609)
- Kepler produces first kinematic description of orbits
- Each object moves in an elliptical orbit
- Newtonian Physics ("Classical Physics" (1687)
- Newton provides dynamical description of universe, using:

1. acceleration
2. momentum
3. conservation laws

[Jed description: Everything on an xyz axis in space-time]

- Quantum Physics ("Modern Physics) (1931)

- Particles are waves and vice-versa
- Probability is all we ever know
- Measurement determines reality

[Jed description: Everything is probabilities]

- _____? (“Post-Modern” Physics)(_____)

COSMOLOGY

The continual iconoclastic evolution in our cosmological understanding of the universe …

- Our Earth is round not flat (this notion of a round earth is contrary to intuition and most of our everyday experience—i.e. over 1/2 of us are hanging upside on this thing) - 500 B.C. (Pythagorus) [Eratosthenes calculated circumference in 240 B.C. and of course Columbus didn’t sail off the edge of the earth when he tried it almost 2000 years later]
- Our Earth isn’t the center of the universe[earth rotates around the sun] - 240 B.C. (Aristarchus) [Calculations by Copernicus in 1532 and astronomical observations by Galileo in 1632]
- Our sun isn’t the center of the universe[stars are other suns] - 450 B.C. (Anaxagoras) [Bruno in 1584 and then Angelo Secchi proved through spectroscopy in 1860]
- Our galaxy isn’t the center of the universe - 964 (Azophi) [Later Kant, Messier, Shapley, and finally Edwin Hubble … January 1, 1925”The day we discovered the universe”]

- Our universe isn't the center of the Universe [or Multiverse]- _____?

---"As usual, nature's imagination far surpasses our own, as we have seen from the other theories which are subtle and deep." - Richard Feynman

---"I think physics is about done with reductionism. It's taken us a long way, but the next jump in physics I think will unleash much higher levels in imaginative heuristics. And then look out universe! Here comes humanity!"- Jed Anderson

---

Licensed CC-BY-4.0 (https://creativecommons.org/licenses/by/4.0/).

Original source: Constant Contact campaign

Markdown source: https://jedanderson.org/posts/it-aint-just-physical.md (https://jedanderson.org/posts/it-aint-just-physical.md)

Source on GitHub: /src/content/posts/it-aint-just-physical.md (https://github.com/jedanderson432/jedanderson-site/blob/main/src/content/posts/it-aint-just-physical.md)